

1. What ensures clear understanding and better collaboration?
 - a) It reduces teamwork
 - b) It makes communication difficult
 - c) It ensures clear understanding**
 - d) It reduces confusion
2. How do you perform basic input validation in JavaScript?
 - a) Using if-else statements**
 - b) Using switch statements
 - c) Using alert() and prompt()
 - d) Using try-catch blocks
3. How do you concatenate strings in JavaScript?
 - a) Using + operator**
 - b) Using * operator
 - c) Using & operator
 - d) Using - operator
4. What is Bootstrap and what role does it play in web design?
 - a) A JavaScript framework
 - b) A CSS framework for building responsive and mobile-first websites**
 - c) A database management system
 - d) A server-side scripting language
5. What is Axios and what is its main purpose?
 - a) A CSS framework
 - b) A cross-platform HTTP client for making HTTP requests**
 - c) A JavaScript library for DOM manipulation
 - d) A database management tool
6. How can you troubleshoot and debug issues related to JavaScript libraries in a web application?
 - a) By using print statements
 - b) By using browser developer tools, console logs, and debugging tools like breakpoints and stack traces**
 - c) By restarting the browser
 - d) By reinstalling the libraries
7. **Which software is best for creating documents?**
 - a) Microsoft Word**
 - b) Adobe Photoshop
 - c) VLC Media Player
 - d) Google Chrome
8. **What is the primary use of Microsoft Excel?**
 - a) Editing videos
 - b) Creating spreadsheets**
 - c) Browsing the internet
 - d) Playing games
9. **What is the purpose of the 'Track Changes' feature in Microsoft Word?**
 - a) To count words
 - b) To review and manage document edits**

- c) To edit images
 - d) To change font size
10. **Which tool in MS Word is used to create a list?**
- a) Bold
 - b) Bullet points**
 - c) Text Alignment
 - d) Font Size
11. **Which of these is NOT a primary color in RGB?**
- a) Red
 - b) Blue
 - c) Yellow**
 - d) Green
12. **Which tool is used to remove a background in Photoshop?**
- a) Clone Stamp
 - b) Magic Wand**
 - c) Dodge Tool
 - d) Healing Brush
13. **What is the function of the Pen Tool in Adobe Illustrator?**
- a) To paint freehand
 - b) To create precise vector paths and shapes**
 - c) To erase elements
 - d) To adjust brightness
14. **Which of these is NOT part of workplace safety?**
- a) Following safety protocols
 - b) Using PPE
 - c) Ignoring minor hazards**
 - d) Reporting unsafe conditions
15. **What is the most effective way to communicate in a professional setting?**
- a) Using informal language
 - b) Being clear and concise**
 - c) Using slang and abbreviations
 - d) Avoiding eye contact
- 16. Why is active listening important in workplace interaction?**
- a) It helps in avoiding work
 - b) It ensures clear understanding and better collaboration

16. What is the purpose of loops in JavaScript?

JavaScript-এ loop ব্যবহার করা হয় কোনো কাজ বারবার করতে
Condition সত্য থাকলে loop চলতে থাকে।

17. How do you handle conditional statements in JavaScript?

JavaScript-এ conditional statement ব্যবহারের জন্য if, else if, else এবং switch ব্যবহার করা হয়।

18. What is the Document Object Model (DOM) in JavaScript?

JavaScript এই DOM ব্যবহার করে ওয়েবপেজের element, style, content ইত্যাদি পরিবর্তন করতে পারে।

19. What is the concept of event bubbling and event capturing in JavaScript?

Event Bubbling:

ইভেন্টটি প্রথমে সবচেয়ে ভেতরের (child) element থেকে শুরু হয়ে বাইরে parent elements-এর দিকে ধাপে ধাপে উপরে উঠে যায়।

• **Event Capturing:**

ইভেন্টটি প্রথমে সবচেয়ে বাইরের (parent) element থেকে শুরু হয়ে ধাপে ধাপে ভেতরের (child) element-এর দিকে নেমে আসে।

20. What role do search engines play in indexing and retrieving information from the Internet?

সার্চ ইঞ্জিন ইন্টারনেটের কোটি কোটি ওয়েবপেজ স্ক্যান করে (crawl) এবং তাদের একটি ডাটাবেসে সংরক্ষণ করে (index)। পরে ব্যবহারকারী কোনো কীওয়ার্ড সার্চ করলে সেই index থেকে সবচেয়ে প্রাসঙ্গিক তথ্য বের করে (retrieve) ফলাফল দেখায়।

21. What are RESTful APIs?

RESTful API হলো একটি ওয়েব সার্ভিস আর্কিটেকচার, যেখানে ক্লায়েন্ট এবং সার্ভার HTTP methods (GET, POST, PUT, DELETE ইত্যাদি) ব্যবহার করে একে অপরের সাথে যোগাযোগ করে। RESTful API সাধারণত JSON ডাটা আদান-প্রদান করে এবং সহজ, দ্রুত ও স্কেলেবল সিস্টেম তৈরি করতে সাহায্য করে।

22. What is PHPMyAdmin, and how is it used in web development?

PHPMyAdmin হলো একটি ওয়েব-ভিত্তিক টুল যা MySQL ডাটাবেস পরিচালনা করতে ব্যবহৃত হয়। এটি ব্যবহার করে ডাটাবেস তৈরি, টেবিল তৈরি, ডাটা ইনসার্ট/ডিলিট, SQL কোয়েরি চালানো এবং ডাটাবেস ব্যাকআপ/রিস্টোর করা যায়। ওয়েব ডেভেলপমেন্টে ডাটাবেস সহজে ম্যানেজ করার জন্য এটি খুবই জনপ্রিয়।

23. How does a static website differ from a dynamic website?

- **Static** → fixed content, simple, fast, low interactivity.
- **Dynamic** → changing content, complex, interactive, personalized.

24. What are some popular frameworks and platforms used for developing dynamic websites?

Laravel, CodeIgniter, Symfony, Express.js, Next.js, Django

25. What are the main features and functionalities of an interactive website?

User Interaction, Dynamic Content, Multimedia Support, Easy Navigation, Responsive Design

26. How do you ensure your designs are compatible with both desktop and mobile browsers?

রেসপন্সিভ, মোবাইল-ফার্স্ট, ফ্লেক্সিবল এবং টেস্টেড ডিজাইন” থাকলেই সব ডিভাইসে কাজ করবে।

27. Write down the name of some key features you look for in a code editor for web design and development tasks.

Web Preview, Git, Syntax Highlighting, Auto-completion, File Browser

28. Create Function using javascript with Example.

```
function sayHello() {  
    console.log("Hello, World!");  
}
```

```
sayHello();
```

29. What are array and object in javascript

An array is a special variable that can hold **multiple values** in a single variable.

An object is a collection of **key-value pairs**. It stores data in a structured way.

Questions with Answers

1. What is the difference between HTML tag, element and attribute?

HTML Tag হলো একটি কোডের অংশ যা ওয়েবপেজে কন্টেন্ট কীভাবে প্রদর্শন হবে তা ব্রাউজারকে নির্দেশ করে। Example - <p> একটা tag

Element – ট্যাগের opening part এবং closing Part এর মাঝখানে যা থাকে তাই Element

Example - <p>Hello NSDA</p> এখানে 'Hello NSDA' Element

Attribute – Tag এর অতিরিক্ত কিছু নির্দেশ করে।

Example <p style="color:red">Hello NSDA</p>

এখানে style একটা Attribute

2. What are the two types of HTML elements and how they function?

Block Level Element - <div> একটা block level element. এটা ব্যবহার করার কারণে content নতুন লাইনে শুরু করবে।

Inline element - একটা inline element. এটা ব্যবহার করলে নতুন লাইনে শুরু হবে না। যত টুকু কন্টেন্ট ঠিক তত টুকু জায়গা নিয়েই থাকবে।

3. Which code editor you use for most of your project and why.?

I use Visual Studio Code because it offers integrated terminal, and Git support.

4. Write the HTML boilerplate.

```
<!DOCTYPE html>
<html>
<head>
  <title>Title</title>
</head>
<body>

</body>
</html>
```

5.How many type of layout Bootstrap have?

Two types: Grid and Flex

6.What does the Bootstrap 'card' class do?

Bootstrap-এর `card` ক্লাস একটি **ফ্লেক্সিবল (flexible)** এবং **এক্সটেনসিবল (extensible)** কন্টেন্ট কনটেইনার তৈরি করে। এটি হেডার, ফুটার, ইমেজ এবং কন্টেন্ট সুন্দরভাবে প্রদর্শনের জন্য ব্যবহার হয়।

7.Difference between JavaScript and Java

Java is a general-purpose, object-oriented programming language,
JavaScript is a scripting language used for interactive web content.

8. Declare a function in JavaScript.

```
function myFunction() {  
    console.log('Hello World');  
}
```

9. What is used to write conditions in JavaScript?

If-else, switch case, ternary operator

10. What is the difference between static, interactive and dynamic website?

Static: fixed content

Interactive: responds to user input,

Dynamic: content generated in real-time using databases.

11. Write the differences between the for loop, while loop, and do-while loop in JavaScript.

For Loop – একটা নির্দিষ্ট নাম্বার পর্যন্ত কোড Execute হবে ।

While loop – Condition True হলে কোড Execute হবে ।

Do-while – প্রথমে একবার execute হবে, এর পর condition চেক করবে ।

12. The most basic element of any HTML page is

ASCII text

13. Write the differences between the DOM and the BOM.

DOM হলো একটি ওয়েবপেজের HTML অথবা XML ডকুমেন্টের গঠন বা কাঠামোর একটি উপস্থাপনা।

BOM হলো ব্রাউজারের উপাদানগুলোর একটি অবজেক্ট মডেল, যার মাধ্যমে JavaScript দিয়ে ব্রাউজার সম্পর্কিত বিভিন্ন কাজ করা যায়

14. Write the HTML boilerplate code.

```
<!DOCTYPE html>
<html>
<head>
  <meta charset="UTF-8">
  <title>Document</title>
</head>
<body>

</body>
</html>
```

15. Create a JSON array for 4 students, including their names, phone numbers, and email addresses.

```
[
  {"name": "Alice", "phone": "1234567890", "email": "alice@example.com"},
  {"name": "Bob", "phone": "2345678901", "email": "bob@example.com"},
  {"name": "Charlie", "phone": "3456789012", "email": "charlie@example.com"},
  {"name": "Diana", "phone": "4567890123", "email": "diana@example.com"}
]
```

16. What is a function in JavaScript, and how does it work? Give an example.

A function is a block of code designed to perform a particular task. It is executed when called.

Example:

```
function greet() {
  return 'Hello';
}
```

17. Write a CSS comment.

```
/* This is a CSS comment */
```

18. Write a JavaScript array.

```
let fruits = ['Apple', 'Banana', 'Cherry'];
```

19. What are margin and padding in CSS?

Margin: border এর বাহিরে space create করে ।

Padding: Content এবং border এর মাঝখানে space create করে ।

Web Design & Development for Freelancing Level 3

Suggestion

1. What are the most common communication Channels used for effective communication?

Ans: Face-to-face, Phone calls, social media

2. What is meant by a website?

Ans: এক বা একাধিক পেজের সমন্বয়ে একটি ওয়েবসাইট তৈরি হয়। ওয়েব সাইটে প্রতিষ্ঠানের তথ্য থাকে।

3. What is meant by Internet Protocol (IP address)?

Ans: প্রত্যেকটি Device এর Unique Address

4. How does computer Hardware differ from software?

Ans: হার্ডওয়্যার - লজিক্যাল ইন্সট্রাকশন এর দরকার হয় না।
যেমন- মাউস একটি হার্ডওয়্যার।

সফটওয়্যার - লজিক্যাল ইন্সট্রাকশন এর দরকার হয়।

যেমন – গেইম

5. What is an interactive website?

Ans: A website that communicates and allows for interaction with users.

6. When does a web developer's personal computer will be called a local server?

Ans: যখন web developer তার নিজের কম্পিউটারে Xampp Server Install করে কাজ করে, তখন সেটাকে Local Server বলে।

যেমনঃ XAMPP একটি লোকাল সার্ভার

7. What is the function of FTP client?

Ans: দুইটি কম্পিউটারের মধ্যে সংযোগ স্থাপন করে এবং ফাইল upload, Download এর জন্য ব্যবহার করা হয়।

8. Write Down the correct sequence of HTML tags for starting a webpage.

Ans: HTML, Head, Title, Body

9. Write the tag used for inserting the largest heading in HTML

Ans: <h1>

10. Write the tag used for inserting a line-break in HTML

Ans:

11. How you will create a hyperlink in HTML?

Ans: Facebook

12. What is meant by cascading style sheets?

Ans: HTML Structure কে সুন্দর ভাবে Design করার জন্য CSS ব্যবহার করা হয়।

13. Write the correct syntax for referring the external style sheet.

Ans: <Link rel="stylesheet" type="text/css" href="style.css">

Web Design & Development for Freelancing Level 3

Suggestion

14. Which color mode is used for printing?

Ans: CMYK

15. Why A block statement is Used in Java script?

Ans: A block statement is generally used to group together several statements.

16. When interpreter encounters an empty statement, what it will do:

Ans: a) Shows a warning

b) Ignores the statements

17. The “function” and “var” are known as:

a) Keywords

b) Data types

c) Declaration statements

18. Write the code in Javascript for getting an output 1.

Ans: `Console.log(1);` অথবা `alert(1);`

19. What will happen, if the following Javascript code is executed?

`Var count = 0;`

`While(count < 10)`

`{`

`Console.log(count)`

`Count++;`

`}`

Ans: it will log the numbers from 0 to 9 to the console

20. List the data type supported by javascript

Ans: 1. Undefined

2. Null

3. Boolean

4. String

5. Symbol

6. Number

7. Object

1 Explain the difference between the == and === operators in JavaScript.

- A) == compares values; === compares values and types
- B) == compares values and types; === compares only values
- C) == compares objects; === compares arrays
- D) == compares strings; === compares numbers

2 How do you ensure accessibility compliance when integrating JavaScript libraries into web applications?

- A) By ignoring accessibility guidelines
- B) By following web accessibility standards, using ARIA roles, and testing with screen readers and other assistive technologies
- C) By using only plain HTML
- D) By minimizing the use of JavaScript

3 What are single-line and multi-line comments in JavaScript? How do you comment out code?

- A) // for single-line, /* */ for multi-line
- B) <!-- --> for single-line, <!-- --> for multi-line
- C) # for single-line, “” ”” for multi-line
- D) // for single-line, /**/ for multi-line

4 Which image format supports transparency and is commonly used for web graphics and images with transparent backgrounds?

- A) PSD
- B) PNG
- C) JPG
- D) GIF

5 What are logical operators?

- A) + - * / B) &&, ||, ! C) ==, !=, ===, !== D) if, else, switch

6 Select the basic data types in JavaScript

- A) Number, String, Boolean, Object, Array
- B) Number, String, Boolean, Undefined, Null, Object
- C) Integer, Float, String, Boolean, Object
- D) Char, String, Boolean, Object, Array

7 How do you use the break statement in loops and switches in JavaScript?

- A) To pause execution
- B) To terminate a loop or switch
- C) To skip an iteration
- D) To continue execution

8 What are arithmetic Operators in JavaScript? Provide examples.

- A) +, -, *, / B) +, -, *, /, &&, ||
- C) ++, --, +=, -= D) =, ===, !=, !==

9 How do you display output in JavaScript?

- A) Using print() B) Using alert()
- C) Using console.log() D) Using document.write()

10 How do you check the data type of a variable in JavaScript?

- A) typeof variable
- B) instanceof variable
- C) variable.type D) check Typeeantiog

11 Which keyword is used to declare a variable in JavaScript?

- a) var
- b) variable c)v d) let

12 Which built-in method is used to convert a string to an integer in JavaScript?

- a) parseInt() b) toInteger()
- c) convertToInt() d) stringToInt()

13 Which operator is used for strict equality comparison in JavaScript?

- a) == b) != c) === d) !==

14 What is the purpose of Tailwind CSS and how does it differ from other CSS frameworks like Bootstrap?

- A) Tailwind CSS is a JavaScript framework
- B) Tailwind CSS is a utility-first CSS framework that provides low-level utility classes, while Bootstrap provides pre-designed components.
- C) Tailwind CSS is used for server-side scripting
- D) Tailwind CSS is a database management tool

15 Which function is used to output data to the console in JavaScript?

a) print() b) log() c) display() d) console.log()

16 Can you explain the basic purpose of FTP (File Transfer Protocol) software in web development?

17 How do you handle conditional statements in JavaScript?

18 How do you create HTML login form with submit button?

19 How does an interactive website differ from static and dynamic websites?

20 Which code editor do you prefer for web design and development projects, and why?

21 What is the difference between innerText and innerHTML in JavaScript?

22 When should I consider using SVG images?

23 How do you create functions in JavaScript?

24 How do you access arrays and objects and their elements/properties?

25 How do you create an ordered list in HTML?

26 What is an ID selector in CSS?

27 Can you explain the difference between for loop, while loop, and do-while loop?

28 How do you apply multiple CSS classes to a single HTML element?

29 What are arrays and objects in JavaScript?

30 What is the Document Object Model (DOM) in JavaScript?

Name:

Phone:

1. কোন ইমেজ ফরম্যাটটি ট্রান্সপারেন্সি সাপোর্ট করে?
A) JPG B) PNG C) GIF D) PSD
2. HTML-এ কোন ট্যাগ দিয়ে ছবি দেখানো হয়?
A) img B) image C) picture D) src
3. JavaScript-এ strict equality operator কোনটি?
A) == B) != C) === D) !==
4. নিচের কোনটি logical operator?
A) + - * / B) && || ! C) == != D) if else
5. break স্টেটমেন্টের কাজ কী?
A) Pause করা B) Loop বন্ধ করা C) Skip করা D) Compile করা
6. JavaScript-এ output দেখাতে কোনটি ব্যবহার করা হয়?
A) print() B) alert() C) console.log() D) document.write()
7. কোনটি JavaScript-এর basic data type নয়?
A) Number B) Boolean C) Undefined D) Char
8. কোন keyword দিয়ে variable declare করা যায়?
A) var B) let C) const D) সবগুলোই
9. কোন মেথড string কে integer-এ convert করতে ব্যবহার হয়?
A) parseInt() B) toInteger() C) convertInt() D) stringToInt()
10. Tailwind CSS কোন ধরনের CSS framework?
A) Component-based B) Utility-first C) Server-side D) Database tool
11. CSS box model-এর কোন অংশ transparent?
A) Padding B) Margin C) Padding & Margin D) Content
12. Bootstrap-এ responsive navbar তৈরি করতে কোন ক্লাস ব্যবহার করা হয়?
A) navbar-light B) navbar-collapse C) navbar-expand D) navbar-fixed
13. SVG কখন ব্যবহার করা উচিত?
A) High-quality scalable graphics B) Photo editing C) Video rendering D) Database design
14. JavaScript-এ কোন লুপ কমপক্ষে একবার চলবেই?
A) for B) while C) do-while D) foreach
15. CSS-এ ID selector কীভাবে লেখা হয়?
A) .id B) #id C) id: D) id=
16. JavaScript-এ == এবং === এর মধ্যে পার্থক্য লিখুন।

17. JS-এ single-line এবং multi-line comment কীভাবে লিখতে হয়?

18. JavaScript-এর basic data types লিখুন।

19. JavaScript-এ variable-এর data type কীভাবে check করা যায়?

20. Arithmetic operators উদাহরণসহ লিখুন।

21. Tailwind CSS এবং Bootstrap-এর মধ্যে মূল পার্থক্য লিখুন।

22. innerText এবং innerHTML এর পার্থক্য লিখুন।

23. Static, Dynamic এবং Interactive website এর পার্থক্য ব্যাখ্যা করুন।

24. FTP software এর কাজ কী?

25. একটি HTML login form-এর basic structure লিখুন।

26. SVG image কখন ব্যবহার করা উচিত?

27. JavaScript-এ কীভাবে function তৈরি করা হয় (কমপক্ষে দুইভাবে)?

28. HTML-এ ordered list কীভাবে তৈরি করা হয়?

29. JavaScript-এ array ও object কী? উদাহরণ দিন।

30. DOM কী? JavaScript কীভাবে DOM manipulate করতে পারে, লিখুন।

1 Which protocol is used by the browser to request a web page?

Answer (English): HTTP or HTTPS protocol.

Answer (Bangla): ওয়েব পেজ রিকোয়েস্ট করার জন্য ব্রাউজার HTTP বা HTTPS প্রোটোকল ব্যবহার করে।

2 Which HTML element provides a container for navigation links?

Answer (English): The <nav> element.

Answer (Bangla): <nav> ট্যাগটি নেভিগেশন লিঙ্কের জন্য কনটেইনার হিসেবে ব্যবহৃত হয়।

3 What is the semantic difference between and tags?

Answer (English): indicates importance; only makes text bold.

Answer (Bangla): শব্দটি গুরুত্বপূর্ণ বোঝায়, আর শুধু টেক্সটটিকে বোল্ড করে।

4 Which CSS property allows a flex item to shrink?

Answer (English): flex-shrink property.

Answer (Bangla): flex-shrink প্রপার্টি আইটেমকে ছোট হতে দেয়।

5 What is the purpose of the repeat() function in CSS Grid?

Answer (English): It defines repeating columns or rows.

Answer (Bangla): এটি গ্রিডের কলাম বা রো পুনরাবৃত্তি করতে ব্যবহৃত হয়।

6 What advantages do frameworks provide in web development?

Answer (English): Faster development, consistent design, and responsive layout.

Answer (Bangla): দ্রুত ডেভেলপমেন্ট, একরূপ ডিজাইন, এবং রেসপনসিভ লেআউট তৈরি সহজ হয়।

7 Which factor affects both image quality and loading speed during optimization?

Answer (English): Image size and file format.

Answer (Bangla): ইমেজের সাইজ ও ফাইল ফরম্যাট দুটোই কোয়ালিটি ও লোডিং স্পিড প্রভাবিত করে।

8 Which selector is most expensive for the browser to render?

Answer (English): Universal selector (*).

Answer (Bangla): ইউনিভার্সাল সিলেক্টর (*) ব্রাউজারের জন্য সবচেয়ে ব্যয়বহুল।

9 Write a coding question based on general JavaScript logic.

Example: Write a program to find the largest number in an array.

উদাহরণ (Bangla): একটি অ্যারে থেকে সবচেয়ে বড় সংখ্যাটি বের করার প্রোগ্রাম লেখো।

10 Write a coding question based on an async function named `fetchData()`.

Example: Create an async function `fetchData()` that fetches data from an API and logs it to the console.

উদাহরণ (Bangla): একটি `fetchData()` async ফাংশন লেখো, যা API থেকে ডেটা এনে কনসোলে দেখাবে।

1 1 What is the ICT Code of Conduct in the workplace?

Answer (English): It's a set of rules for responsible, ethical, and professional ICT use at work.

Answer (Bangla): এটি কর্মস্থলে দায়িত্বশীল ও নৈতিক ICT ব্যবহারের জন্য নিয়মাবলি।

1 2 What is Mail Merge in a word processor?

Answer (English): It's used to send personalized letters or emails to many people at once.

Answer (Bangla): এটি একই ডকুমেন্টকে অনেকজনের জন্য আলাদা আলাদা নামে পাঠাতে ব্যবহৃত হয়।

1 3 What is the basic folder structure for a web design project?

Answer (English):

project/

├— index.html

├— css/

├— js/

└— images/

Answer (Bangla):

ওয়েব প্রজেক্টে সাধারণত HTML, CSS, JS, এবং images ফোল্ডার থাকে।

1 4 What is the difference between <section>, <div>, and <article>?

Answer (English):

- <section> = thematic group of content
- <article> = self-contained content
- <div> = general container

Answer (Bangla):

- <section> = বিষয়ভিত্তিক অংশ
- <article> = স্বতন্ত্র কন্টেন্ট
- <div> = সাধারণ কন্টেইনার

1 5 What is the difference between raster and vector images?

Answer (English):

Raster = pixel-based; Vector = shape-based and scalable.

Answer (Bangla):

রাস্টার ইমেজ পিক্সেলভিত্তিক, ভেক্টর ইমেজ আকারভিত্তিক ও বড় করলে ভাঙে না।

1 6 What is the difference between relative and absolute units in CSS?

Answer (English):

Relative units depend on parent size (like em, %); absolute units (like px, cm) are fixed.

Answer (Bangla):

রিলেটিভ ইউনিট প্যারেন্টের সাইজের উপর নির্ভর করে, আর অ্যাবসলিউট ইউনিট স্থির থাকে।

1 7 What is an array, and how is forEach() used?

Answer (English):

Array stores multiple values; forEach() loops through each element.

Answer (Bangla):

অ্যারে একাধিক মান রাখে, forEach() প্রতিটি মানের উপর লুপ চালায়।

1 8 What is the difference between addEventListener() and onclick?

Answer (English):

addEventListener() allows multiple events; onclick only one.

Answer (Bangla):

addEventListener() একাধিক ইভেন্ট হ্যান্ডেল করতে পারে, onclick একটাই পারে।

1 9 What is the difference between forEach() and map()?

Answer (English):

forEach() executes code only; map() returns a new array.

Answer (Bangla):

forEach() শুধু কাজ করে, map() নতুন অ্যারে রিটার্ন করে।

2 0 What is the purpose of the z-index property?

Answer (English):

It controls the stacking order of elements.

Answer (Bangla):

এটি এলিমেন্টগুলোর ওপরে-নিচে অবস্থান নিয়ন্ত্রণ করে।

2 1 How does Flexbox help in building a responsive website?

Answer (English):

It automatically adjusts item size and position based on screen width.

Answer (Bangla):

ফ্লেক্স সাইজ অনুযায়ী আইটেমের অবস্থান ও সাইজ পরিবর্তন করতে সাহায্য করে।

2 2 What are the advantages of CSS frameworks over custom CSS?

Answer (English):

Faster development, built-in responsiveness, and cross-browser support.

Answer (Bangla):

দ্রুত কাজ করা যায়, রেসপনসিভ ডিজাইন সহজ হয়, সব ব্রাউজারে ভালো কাজ করে।

2 3 What are the different CSS position values and their purposes?

Answer (English):

static, relative, absolute, fixed, sticky.

Answer (Bangla):

CSS পজিশন মানগুলো হলো: static, relative, absolute, fixed, sticky।

2 4 What is the difference between == and === in JavaScript?

Answer (English):

== compares value only; === compares value and type both.

Answer (Bangla):

== শুধু মান তুলনা করে, === মান ও টাইপ দুটোই তুলনা করে।

2 5 What is the main purpose of the DOM (Document Object Model)?

Answer (English):

It allows JavaScript to access and manipulate HTML elements.

Answer (Bangla):

DOM এর মাধ্যমে JavaScript HTML এলিমেন্ট নিয়ন্ত্রণ করতে পারে।

2 6 Give an example of using for...of with an array.

Example (English):

```
let fruits = ["apple", "banana", "mango"];  
for (let f of fruits) {  
  console.log(f);  
}
```

উদাহরণ (Bangla):

এই লুপ প্রতিটি ফলের নাম কনসোলে দেখাবে।

2 7 What is the use of try...catch in JavaScript?

Answer (English):

To handle errors without stopping the program.

Answer (Bangla):

কোনো এরর হলে প্রোগ্রাম বন্ধ না করে সেটি হ্যান্ডেল করতে ব্যবহৃত হয়।

2 8 What is the bind() method used for in JavaScript?

Answer (English):

It binds a function to a specific object's this value.

Answer (Bangla):

bind() কোনো ফাংশনের this নির্দিষ্ট অবজেক্টের সাথে যুক্ত করে।

2 9 What are media queries, and how are they used for responsiveness?

Answer (English):

Media queries apply different CSS styles based on device width.

Answer (Bangla):

মিডিয়া কুয়েরি ডিভাইসের প্রস্থ অনুযায়ী ভিন্ন CSS স্টাইল প্রয়োগ করে।

3 0 What is the difference between synchronous and asynchronous operations in JS?

Answer (English):

Synchronous = one task at a time; Asynchronous = multiple tasks run without waiting.

Answer (Bangla):

সিঙ্ক্রোনাস একসময় এক কাজ করে, অ্যাসিঙ্ক্রোনাস একাধিক কাজ একসাথে চালায়।

Name:

Phone:

1. What is the difference between HTML tag, element and attribute?
2. What are the two types of HTML elements and how they function?
3. Which code editor you use for most of your project and why.?
4. Write the HTML boilerplate.
5. How many type of layout Bootstrap have?
6. What does the Bootstrap 'card' class do?
7. Difference between JavaScript and Java
8. What is used to write conditions in JavaScript?

9. What is the difference between static, interactive and dynamic website?
10. Write the differences between the for loop, while loop, and do-while loop in JavaScript.
11. The most basic element of any HTML page is
12. Write the differences between the DOM and the BOM.
13. Create a JSON array for 4 students, including their names, phone numbers, and email addresses.

14. What is a function in JavaScript, and how does it work? Give an example.

17. Write a CSS comment.

18. Write a JavaScript array.

19. What are margin and padding in CSS?

16. Which image format is best suited for photographs and images with gradients due to its ability to handle millions of colors and efficient compression?

A) PNG

B) GIF

C) JPG/JPEG

D) PSD

17. Which image format supports transparency and is commonly used for web graphics and images with transparent backgrounds?

A) PSD

(B) PNG

C) JPG

D) GIF

18. What file extension is commonly associated with Photoshop plug-in files?

A) .exe

B) .dll

C) .8bf

D) .psd

19. What is the function of a layer mask?

A) To change the color of a layer

B) To hide or reveal parts of a layer

C) To increase the layer's opacity

D) To duplicate a layer

20. What is a Smart Object?

A) A rasterized layer

B) A Layer that can be edited non destructively

C) A text layer

D) A layer without transparency

21. How do you display output in JavaScript?

A) Using print()

B) Using alert()

C) Using console.log()

D) Using document.write()

22. What are single-line and multi-line comments in JavaScript? How do you comment out code?

A) for single-line, /* */ for multi-line

B) <!--> for single-line, <!-- --> for multi-line

C) # for single-line, "" "" for multi-line

D) // for single-line, /**/ for multi-line

23. Select the basic data types in JavaScript

A) Number, String, Boolean, Object, Array

B) Number, String, Boolean, Undefined, Null, Object

C) Integer, Float, String, Boolean, Object

D) Char, String, Boolean, Object, Array

24. How do you check the data type of a variable in JavaScript?

A) typeof variable

B) instanceof variable

C) variable.type

D) checkType(variable)

25. What are arithmetic operators in JavaScript? Provide examples.

- A) +, -, *, /
- B) +, -, *, /, &&, ||
- C) ++, --, +=, -=
- D) ===, !==, !=, !=

26. Which keyword is used to declare a variable in JavaScript?

- a) var
- b) variable
- c) v
- d) let

27. Which operator is used for strict equality comparison in JavaScript?

- a) ==
- b) !=
- c) ===
- d) !==

28. Which function is used to output data to the console in JavaScript?

- a) print()
- b) log()
- c) display()
- d) console.log()

29. Which built-in method is used to convert a string to an integer in JavaScript?

- a) parseInt()
- b) toInteger()
- c) convertToInt()
- d) stringToInt()

30. What does the method `querySelector()` do in JavaScript?

- a) It selects multiple elements based on a class name.
- (b) It selects the first element based on a CSS selector.
- c) It selects an element based on its ID.
- d) It selects all elements of a specific tag name.

1 How do you generate random numbers in Java Script?

- a) Using `math.random()` c) Using `getrandom()`
- b) Using `random()` d) Using `generate random()`

2 How do you create a multiline string in JavaScript?

- a) Using double quotes **b) Using backticks**
- c) Using parentheses d) Using single quotes

3 What are the `let`, `const`, and `class` keywords introduced in ES6? How do they differ from their ES5 counterparts?

- a) `let` for block-scoped variables, `const` for constants, `class` for classes**
- b) `let` for global variables, `const` for constants, `class` for functions
- c) `let` for functions, `const` for block-scoped variables, `class` for arrays
- d) `let` for objects, `const` for global variables, `class` for constants

4 What is jQuery and what is its main purpose in web development?

- A) A CSS framework
- B) A JavaScript library for simplifying HTML DOM manipulation, event handling, and animation**
- C) A backend framework
- D) A database management system

5 What is the purpose of the "alt" attribute in HTML image tags?

- a) It specifies the alignment of the image.
- b) It defines the size of the image.
- c) It provides alternative text for the image for accessibility.**
- d) It controls the animation of the image.

6 What is Bootstrap and what role does it play in web design?

- A) A JavaScript framework
- B) A CSS framework for building responsive and mobile-first websites**
- C) A database management system
- D) A server-side scripting language

7 Which CSS property is used to control the spacing between lines of text?

- a) `text-spacing` c) `letter-spacing`
- b) `line-spacing` **d) `line-height`**

8 What does the "float" property in CSS do?

a) It controls the alignment of text within an element.

b) It removes an element from the normal flow of the document and positions it to the left or right of its containing element.

c) It makes an element visible or hidden.

d) It rotates an element clockwise or counterclockwise.

9 What does the "z-index" property in CSS control?

a) It controls the size of an element.

b) It controls the position of an element in the stacking order.

c) It controls the opacity of an element.

d) It controls the rotation of an element.

10 Which HTML tag is used to define a table row?

a) <tr> c) <th>

b) <td> d) <table>

11 Which keyword is used to declare a variable in JavaScript?

a) var c) v

b) variable d) let

12 Which operator is used for strict equality comparison in JavaScript?

a) == c) ===

b) != d) !=

13 Which function is used to output data to the console in JavaScript?

a) print() c) display()

b) log() **d) console.log()**

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d) string Tolnt() b) toInteger()

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b) It selects the first element based on a CSS selector.

c) It selects an element based on its ID.

d) It selects all elements of a specific tag name.

16 How do VPNs (Virtual Private Networks) enhance intranet security and connectivity?

VPNs encrypt data, making it secure during transmission. They allow remote access to intranet resources by creating a

secure, private connection over the internet.

17 How does an interactive website differ from static and dynamic websites?

Static websites are fixed and show the same content to all users.

Dynamic websites fetch content dynamically from a database based on user interactions.

Interactive websites provide richer engagement, allowing user input and interactive elements like forms, animations,

and real-time updates

18 Which code editor do you prefer for web design and development projects, and why?

Examples include VS Code for its flexibility, extensions, and debugging tools, or Sublime Text for its lightweight performance.

19 Can you explain the basic purpose of FTP (File Transfer Protocol) software in web development?

FTP is used to transfer files between a local computer and a web server, making it essential for uploading and managing

website content.

20 How do you create HTML login form with submit button?

```
<form action="/submit-login" method="POST">  
<label for="username">Username:</label>  
<input type="text" id="username" name="username">  
<label for="password">Password:</label>  
<input type="password" id="password" name="password">  
<button type="submit">Login</button>  
</form>
```

21 When should I consider using SVG images?

Use SVGs for scalable graphics like icons and logos because they maintain quality at any resolution and are lightweight.

22 What is an ID selector in CSS?

An ID selector targets a specific element using its unique ID, prefixed with # in CSS.

23 How do you apply multiple CSS classes to a single HTML element?

```
<div class="class1 class2"></div>
```

24 How do you create functions in JavaScript?

```
function greet(name) {  
  return `Hello, ${name}!`;  
}
```

25 Can you explain the difference between function declarations and function expressions?

Function Declaration: Defined with function keyword and hoisted

Function Expression: Stored in a variable and not hoisted.

26 What are arrays and objects in JavaScript?

Array: Ordered collection of elements.

Example: [1, 2, 3]

Object: Key-value pairs.

Example: { name: "Alice", age: 25 }

27. What are arrays and objects in JavaScript?

Array: Ordered collection of elements.

Example: [1, 2, 3]

Object: Key-value pairs.

Example: { name: "Alice", age: 25 }

28 Can you explain the difference for loop, while loop and do while loop?

For loop: Executes a set number of times

While loop: Runs while a condition is true.

Do-while loop: Executes at least once, then checks the condition.

29 How do you handle conditional statement in java script?

Using if-else, switch, or ternary operators.

30 How do you create new Google Sheet in your Google Drive

Open Google Drive.

Click New > Google Sheets. Name and customize your sheet.